

Keywords: causal discovery • foundation models • benchmark • structural causal models • memorization

CausalArena: Benchmarking Causal Discovery in the Foundation Model Era

Fixed Synthetic Benchmarks Overestimate CDFM Performance by 15–28 Points Due to Pretraining Overlap — Classical Algorithms Remain Competitive on Novel SCMs

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arXiv:2609.11897

Causal discovery has entered the foundation model era: pretrained causal discovery foundation models (CDFMs) learn to predict DAGs in-context, and LLM-based causal agents reason about structure from variable names. Both paradigms achieve impressive scores on existing benchmarks — but these benchmarks are fixed, synthetic, and drawn from the same SCM families that CDFMs were pretrained on. CausalArena introduces a unified, evolvable benchmark that controls for pretraining overlap and enables, for the first time, meaningful cross-paradigm comparisons. On genuinely novel SCMs, classical algorithms match or outperform CDFMs; on semantically rich named-variable graphs, LLM agents dominate. No paradigm is universally best.

AT A GLANCE

Problem: Fixed synthetic benchmarks inflate CDFM performance due to pretraining–test overlap; cross-paradigm comparison is impossible under inconsistent protocols.

Why it matters: Practitioners need to choose the right causal discovery tool; misleading benchmarks produce wrong choices.

Method: Procedural SCM generator producing novel graphs; common protocol with SHD, F1, and new CER metric; evolvable test set refreshed monthly.

Result: CDFMs score 8–14 SHD points worse on out-of-pretraining SCMs; classical algorithms match CDFMs on novel SCMs; LLM agents dominate on named-variable graphs.

Evidence: 11 algorithms across 3 paradigms; 12 SCM families; canary memorization test; 3 data regimes.

Causal discovery is the task of inferring a directed acyclic graph (DAG) $\mathcal{G} = (V, E)$ from observational data $\mathbf{X} \sim P_{\mathcal{G}}$. Three paradigms compete:

Classical algorithms (PC, FCI, GES, NOTEARS) operate directly on data via conditional independence tests or structural score optimization. They have no pretraining.

CDFMs (CausalFormer, DiffAN, CDFL) are pretrained on millions of synthetic SCM-generated datasets and predict DAGs in-context given a new dataset — analogous to in-context learning in language models.

LLM causal agents (GPT-4o-Causal, Claude-3.5-Causal) use semantic knowledge of variable names and domain background to reason about causal relations, augmented by statistical tests.

Each paradigm has been evaluated on different benchmarks with different graph families, mechanisms, and metrics. CDFMs report large wins over classical algorithms on their benchmark — but their benchmark uses the same SCM families they were pretrained on.

The Memorization Problem

The authors prove that CDFMs exhibit significant memorization: their performance on SCM families that appear in their pretraining corpus is 15–28 SHD points better than on structurally novel families.

Memorization gap:

$$\Delta_{\text{mem}} = \text{SHD}(M, \mathcal{G}_{\text{canonical}}) - \text{SHD}(M, \mathcal{G}_{\text{canary}})$$

where $\mathcal{G}_{\text{canary}}$ is a “canary” SCM family constructed by the authors that does not appear in any published CDFM pretraining curriculum. All three CDFMs show $\Delta_{\text{mem}} > 5$ SHD points ($p < 0.001$).

This means published CDFM benchmark numbers are inflated by pretraining memorization, not causal discovery ability.

CausalArena Design

SCM Generator. CausalArena procedurally generates novel SCMs outside any existing CDFM pretraining corpus:

- *Graph families:* Erdős–Rényi, scale-free, bipartite layered, causal motif (new).
- *Mechanisms:* Linear Gaussian, ANM, post-nonlinear, discretized multinomial.
- *Semantics:* Anonymized vs. domain-named variables.
- *Sizes:* 5–50 nodes.

Common Protocol. CausalArena enforces:

The Evaluation Crisis in Causal Discovery

- *Metrics*: SHD, F1 on directed edges, and the new Causal Effect Recovery (CER) metric.
- *Data regimes*: $n \in \{500, 5000, 50000\}$.
- *Knowledge levels*: Zero-knowledge (anonymous), partial (names), full (names + background document).

Evolvability. The test set is seeded by a public random key and refreshed monthly to prevent re-memorization. A version-controlled leaderboard tracks progress over time.

The CER Metric

SHD measures graph structure but not causal effect accuracy. CausalArena introduces **Causal Effect Recovery (CER)**:

$$\text{CER} = 1 - \frac{1}{|P|} \sum_{(i,j) \in P} \frac{|\hat{\tau}_{ij} - \tau_{ij}|}{|\tau_{ij}|}$$

where P is the set of all variable pairs, τ_{ij} is the true total causal effect of X_i on X_j (via *do*-calculus), and $\hat{\tau}_{ij}$ is estimated from the predicted DAG via structural equation fitting. CER directly measures whether the predicted graph supports correct downstream interventional reasoning.

Experimental Findings

Novel SCMs, $n = 5,000$, anonymized variables:

Algorithm	SHD ↓	F1 ↑	CER ↑
GES	7.9	0.64	0.74
PC	8.4	0.61	0.71
CausalFormer	9.8	0.57	0.65
DiffAN	10.3	0.54	0.63
GPT-4o-Causal	18.7	0.31	0.41

Classical algorithms outperform CDFMs on novel, anonymized SCMs — reversing the apparent superiority CDFMs show on fixed benchmarks.

Named variables, $n = 5,000$:

Algorithm	SHD ↓	F1 ↑	CER ↑
GES	8.1	0.63	0.73
CausalFormer	10.1	0.56	0.64
GPT-4o-Causal	6.2	0.74	0.82

LLM agents dominate when variable names carry domain-specific semantics.

Ablations and Interpretation

Sample size: CDFMs show flat performance across $n \in \{500, 5000, 50000\}$ — they rely on learned structure priors. Classical algorithms improve monotonically with n .

Mechanism complexity: Post-nonlinear mechanisms

are hardest for all methods; CDFMs trained on linear Gaussian data show a 12-point SHD gap on nonlinear instances.

Graph size: All algorithms degrade with more nodes; classical algorithms degrade more gracefully at $n > 20$ nodes than CDFMs.

Semantic grounding: LLM agents degrade sharply on anonymized variables (−12.5 SHD points), showing their advantage is entirely domain-knowledge driven.

Reference

Zi-Rong Li, Si-Yang Liu, Tian-Zuo Wang, Han-Jia Ye. **CausalArena: Benchmarking Causal Discovery in the Foundation Model Era.** arXiv:2609.11897, September 2026. <https://arxiv.org/abs/2609.11897>